WebClient Computing

CSS Basics

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Announcements

- First homework is online
  - GitHub - https://github.com/kmu-leeky/kmu-cs-web-client-161/tree/homework-1
  - In slack as a HTML file
  - The submission has to be in the HTML format – write answers in HTML
- GitHub tutorial session at the end of the course from 17:45
  - Fetch and update from remote branches

Using External CSS File is an Ideal Way

- Benefits of external CSS
  - Share a CSS file across different HTML pages (reusable)
  - Reduce HTML file size
    - This can be especially beneficial when the content (HTML) update happens frequent while a CSS update happens rarely
    - A browser can cache CSS file to reduce network overhead
    - Lower loading time (note CSS file can be cached)

Characters of Using Internal CSS File

- Benefits of internal CSS
  - One style of same element – note selector can be tag name, class
  - Might be easier to work as style and HTML is in a sinlge file
- Disadvantage of internal CSS
  - If same style applies to multiple HTML documents, the style has to be copied
  - File size can be large

Characters of Using Inline CSS

- Note that the inline CSS has the highest priority
- Benefits of inline CSS
  - Good for test – can easily try a new feature
- Disadvantages of inline CSS
  - Overriding internal and external CSS file – a page can render in an unexpected way
  - Has to be applied for every element

CSS Border Attribute Shorthand

- To set border,
  - border-width
  - border-style
  - border-color
- We can shorten it as
  - border: <width> <style> <color>

Border Demo

Border Demo

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .border-full {
      border-width: 5px;
      border-style: dotted;
      border-color: blue;
    }
    .border-shorthand {
      border: 5px dotted blue;
    }
  </style>
</head>
<body>
  <p class="border-full">Border Demo</p>
  <p class="border-shorthand">Border Demo</p>
</body>
</html>
```

Block Element and Inline Element

- Block element
  - Always starts in a new line and takes up full width
  - div, h1 ~ h6, p, …
- Inline element
  - Does not start on a new line and takes up only necessary width
  - a, img, span, …

CSS Width and Height

- Using width and height property, we can set the height and width of block elements
- Note the difference between width and max-width

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .width-demo {
      width: 500px;
      height: 100px;
      border: 3px solid #73AD21;
      margin: 10px;
    }
    .max-width-demo {
      max-width: 500px;
      height: 100px;
      border: 3px solid #73AD21;
      margin: 10px;
    }
  </style>
</head>
<body>
  <div class="width-demo">setting width/heigh of a div using width.
    setting width/heigh of a div using width</div>
  <div class="max-width-demo">setting width/heigh of a div using max-width.
    setting width/heigh of a div using max-width.</div>
</body>
</html>
```

Text Styles

- Text alignment – decide where to locate texts
  - text-align: center/left/right/justify
- Text color – decide text color
  - color: blue/red… - http://www.w3schools.com/colors/colors_names.asp
- Text decoration – decoration of a text
  - text-decoration: none/overline/line-through/underline
- Text transformation – specify upper/lowercase in a text
  - text-transform: uppercase/lowercase/capitalize
- Text indentation – specify the indentation of a first line
  - text-indent: 50px

Text Styles

- Word spacing – specify the space between the words
  - word-spacing: 10px;
- Letter spacing – specify the space between the characters
  - letter-spacing: 3px;
- Line Height – space between lines
  - line-height: 3; // unitless – multiplies with the element’s font size

Text Styles Example

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .text-1 {
      color: blue;
      text-align: left;
      text-decoration: overline;
      text-transform: uppercase;
      text-indent: 50px;
      letter-spacing: 3px;
      word-spacing: 10px;
      line-height: 0.8;
    }
    .text-2 {
      text-align: center;
      text-decoration: line-through;
      text-transform: capitalize;
      text-indent: 100px;
      letter-spacing: 10px;
      word-spacing: 20px;
      line-height: 1.6;
    }
  </style>
</head>
<body>
  <div class="text-1">Various text related effects.
  Various text related effects.
  Various text related effects.</div>
  <div class="text-2">Other text effects. Other text effects.
    Other text effects. Other text effects.</div>
</body>
</html>
```

Fonts

- font-family allows to specify fonts to be used in a document
- Multiple fonts can be written in the priority order in CSV format
  - font-family: Arial, “Times New Roman”, “Courier New”, Serif
- Generic family: A set of fonts with similar look. Has to be used as a last fallback mechanism
  - serif : line at the of characters
  - sans-serif : no line at the end of character
  - monospace : all characters have same width
  - cursive
  - fantasy

Fonts Family

- Font Family
  - Under the generic family, more fine grained set of fonts
  - Serif – Times New Roman, Georgia
  - Sans-serif – Arial, Vernada
  - Monospace – Courier New, Lucida Console

Fonts Example

This font in in Serif generic family

This font in in Sans-Serif generic family

This font in in Monospace generic family

This font in in Cursive generic family

This font in in Fantasy generic family

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .serif {
      font-family: Times, "Times New Roman", Georgia, serif;
    }
    .sansserif {
      font-family: Verdana, Arial, Helvetica, sans-serif;
    }
    .monospace {
      font-family: "Lucida Console", Courier, monospace;
    }
    .cursive {
      font-family: cursive;
    }
    .fantasy {
      font-family: fantasy;
    }
  </style>
</head>
<body>
  <div class="serif">This font in in Serif generic family</div>
  <div class="sansserif">This font in in Sans-Serif generic family</div>
  <div class="monospace">This font in in Monospace generic family</div>
  <div class="cursive">This font in in Cursive generic family</div>
  <div class="fantasy">This font in in Fantasy generic family</div>
</body>
</html>
```

Font Effects

- Using font-style, we can have an italic character effect
  - Remember the tag not to use to have an italic character
- Using font-weight, we can make a font bold
- Font size can be set using font-size
  - Unit in px, em, %
  - 1 em = default font size of a browser (generallly 16 px)
  - Percentile is ratio from the default size

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .default {
      font-size: 100%;
      font-style: normal;
      font-weight: normal;
    }
    .effect {
      font-size: 200%;
      font-style: italic;
      font-weight: bold;
    }
  </style>
</head>
<body>
  <p class=default> A paragraph with a default font effect</p>
  <p class=effect> A paragraph with some effects</p>
</body>
</html>
```

A paragraph with a default font effect

A paragraph with some effects

Links Decoration

- Link tag (<a>) has multiple states
  - a:link – an unvisited link
  - a:visited – a visited (clicked) link
  - a:hover – when a mouse is over the link
  - a:active – at the moment when the mouse is clicked
- Each state can have various styles
  - Color, font, background-color, ..
  - text-decoration : decide whether underline is drawn

Kookmin University Computer Science

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    a:link {
      color: red;
      text-decoration: none;
      padding: 14px 25px;
      text-align: center;
      background-color: blue;
    }
    a:visited {
      color: green;
      text-decoration: underline;
    }
    a:hover {
      color: black;
      text-decoration: none;
    }
    a:active {
      color: yellow;
      text-decoration: underline;
    }
    </style>
</head>
<body>
  <a href="http://cs.kookmin.ac.kr">Kookmin University Computer Science</a>
</body>
</html>
```

Lists Decoration

- Remember <ul> and <ol>
  - <ul> Unordered List
  - <ol> Ordered List
- list-style-type allows to set the type of item marker
  - Circle
  - Square
  - A, a
  - I, i

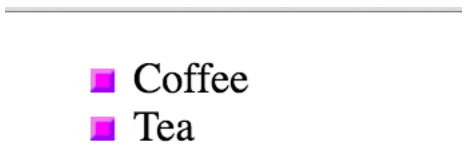
- Coffee
- Tea
- Milk

A. Coffee
B. Tea
C. Milk

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML lists</title>
</head>
<body>
  <ul style="list-style-type:circle">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ul>
  <ol type="A">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ol>
</body>
</html>
```

Lists Decoration

- Using list-style-image to use an image as an item marker
  - list-style-image: url('sqpurple.gif');




```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    ul {
      list-style-image: url('sqpurple.gif');
    }
  </style>
</head>
<body>
  <ul>
    <li>Coffee</li>
    <li>Tea</li>
  </ul>
</body>
</html>
```

Table Borders

- Using border, we can set the shape of border in a table
- border-collapse
  - Use a single line border

Firstname	Lastname
Peter	Griffin
Lois	Griffin

Firstname	Lastname
Peter	Griffin
Lois	Griffin

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    table, th, td {
      border: 1px solid red;
      border-collapse: collapse;
    }
  </style>
</head>
<body>
  <table>
    <tr>
      <th>Firstname</th>
      <th>Lastname</th>
    </tr>
    <tr>
      <td>Peter</td>
      <td>Griffin</td>
    </tr>
    <tr>
      <td>Lois</td>
      <td>Griffin</td>
    </tr>
  </table>
</body>
</html>
```

Table Borders

- Width and Height
  - Sets width and height
- Text-align
  - Sets the horizontal alignment
- Vertical-align
  - Set the vertical alignment

Firstname	Lastname	Savings
Peter	Griffin	$100
Lois	Griffin	$150

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    table, td, th {
      border: 1px solid black;
    }
    table {
      border-collapse: collapse;
      width: 100%;
    }
    th {
      height: 70px;
    }
    td {
      height: 40px;
      text-align: center;
      vertical-align: bottom;
    }
  </style>
</head>
```

Table border-bottom, padding, hover

- Bodder-bottom : draws a line per each line
  - border-bottom: 1px solid gray;
- Padding : space between characters and the cell
  - padding: 10px
- Hover : change color when a mouse pointer is at a row
  - tr:hover {background-color: #f5f5f5}

Example

First Name	Last Name	Points
Peter	Griffin	$100
Lois	Griffin	$150

```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    th, td {
      padding: 10px;
      text-align: left;
      border-bottom: 1px solid gray;
    }
    tr:hover{background-color:gray}
  </style>
</head>
<body>
  <table>
    <tr>
      <th>First Name</th>
      <th>Last Name</th>
      <th>Points</th>
    </tr>
    <tr>
      <td>Peter</td>
      <td>Griffin</td>
      <td>$100</td>
    </tr>
    <tr>
      <td>Lois</td>
      <td>Griffin</td>
      <td>$150</td>
    </tr>
  </table>
</body>
</html>
```

Stripped Table

- Have an effect of stripped colors across different rows (tr)
  - tr:nth-child(even) {background-color: #f2f2f2}

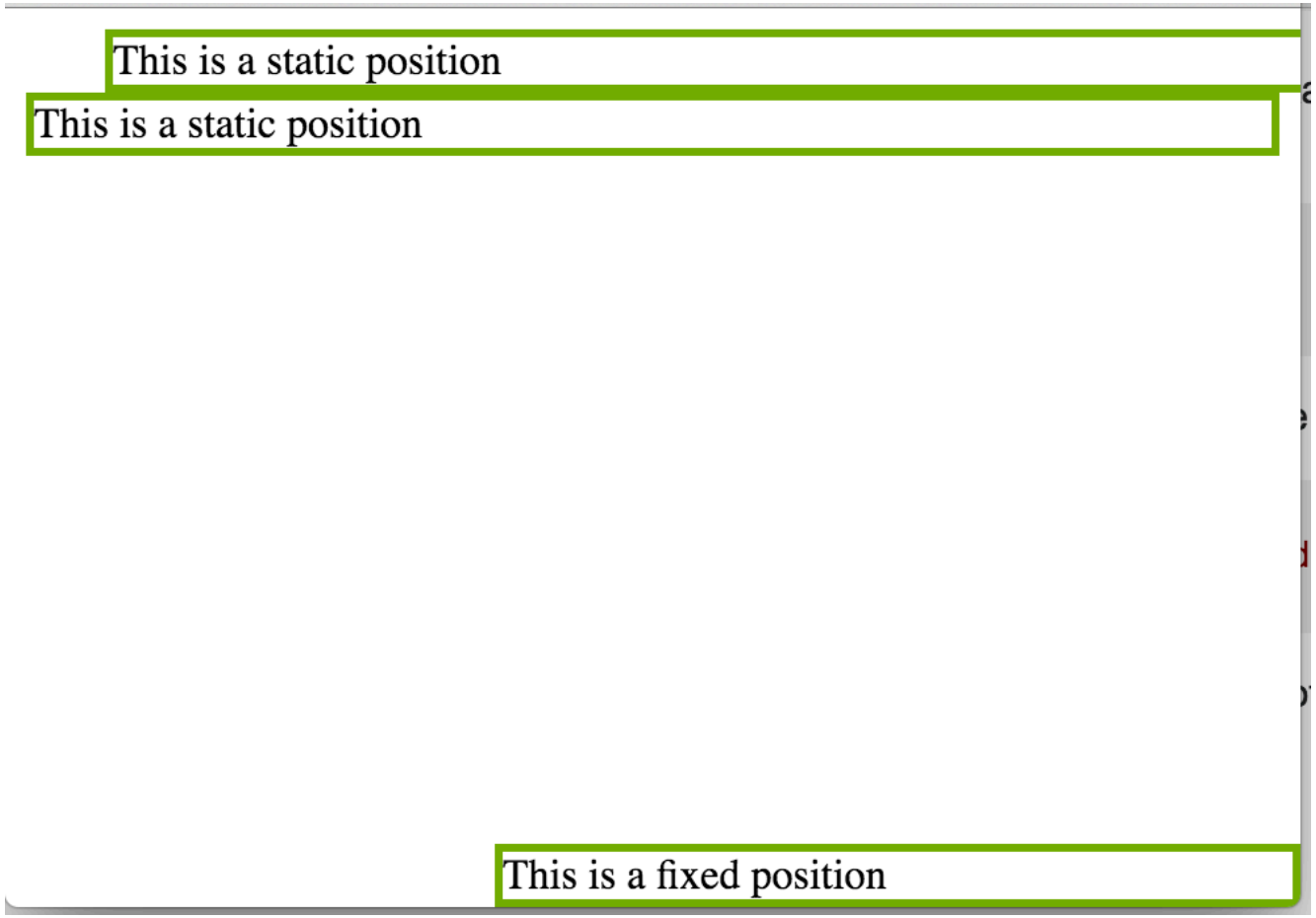
First Name	Last Name	Points
Peter	Griffin	$100
Lois	Griffin	$150
Joe	Swanson	$300
Joe	Swanson	$300

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    th, td {
      padding: 10px;
      text-align: left;
      border-bottom: 1px solid gray;
    }
    th {
      background-color: green;
      color: white;
    }
    tr:nth-child(even) {background-color:gray}
  </style>
</head>
<body>
  <table>
```

CSS Position Property

- Position property sets the location method of elements
  - static, relative, fixed
- Position: static
  - The default location
- Position: relative
  - Position is determined relative to its normal position (static position)
  - top,right, bottom, and left can be set
- Position: fixed
  - Located always at the specified fixed position even with scrolling
  - top,right, bottom, and left can be set

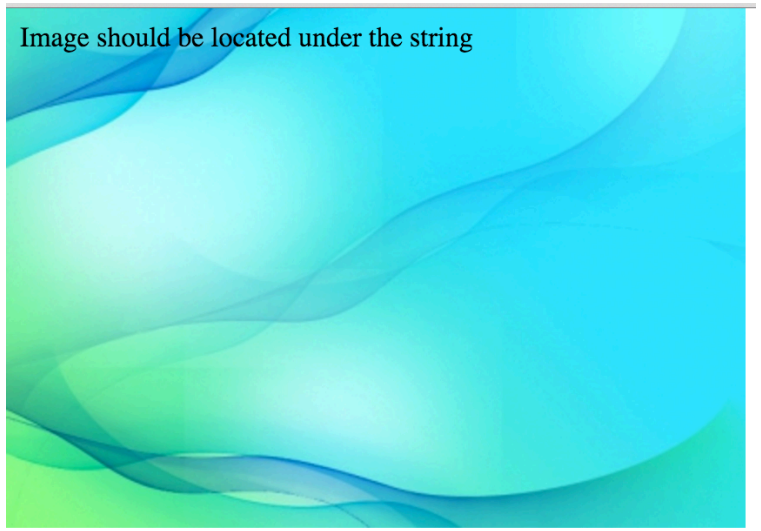
CSS Position Example




```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .static {
      position: static;
      border: 3px solid #73AD21;
    }
    .relative {
      position: relative;
      left: 30px;
      border: 3px solid #73AD21;
    }
    .fixed {
      position: fixed;
      bottom: 0;
      right: 0;
      width: 300px;
      border: 3px solid #73AD21;
    }
  </style>
</head>
<body>
  <div class="relative">This is a static position</div>
  <div class="fixed">This is a fixed position</div>
  <div class="static">This is a static position</div>
</body>
</html>
```

CSS z-index

- It decides whether an element is located on front or back in case of two elements duplicate in a same location
- An element with larger z-index is on top

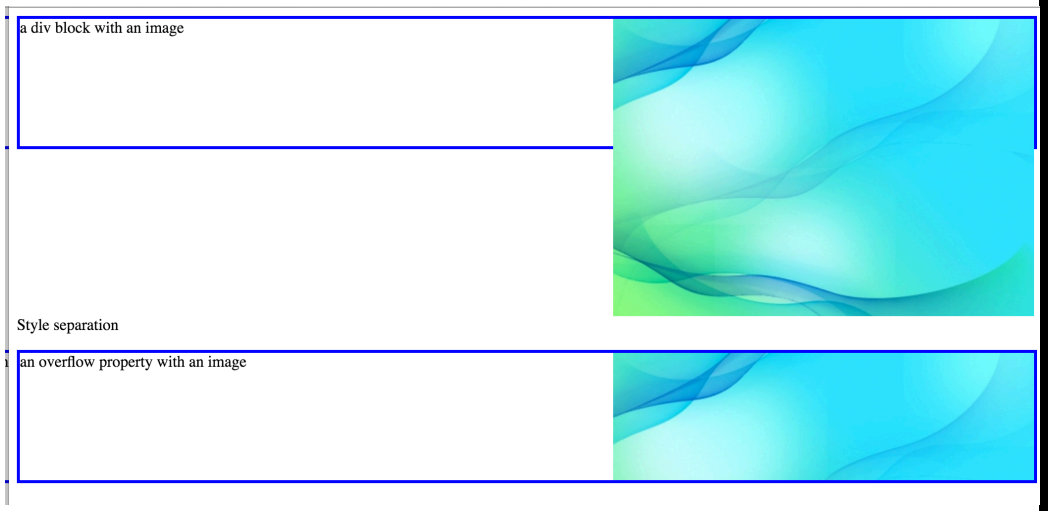



```html
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
  <style>
    .zindex {
      position: fixed;
      left: 0;
      top: 0;
      z-index: -1;
    }
  </style>
</head>
<body>
  Image should be located under the string
  <img class="zindex" src="bg_image.jpg">
</body>
</html>
```

CSS Float and Overflow

- Float property allows to set how an image is located around text
  - left/right
- Overflow property allows to make a containing object to resize based on the size of the internal object
  - auto – automatically set to resize
  - visible – the overflow part is shown (default)
  - hidden – overflow part is not shown
  - scroll – enables a scroll bar for the clipped region

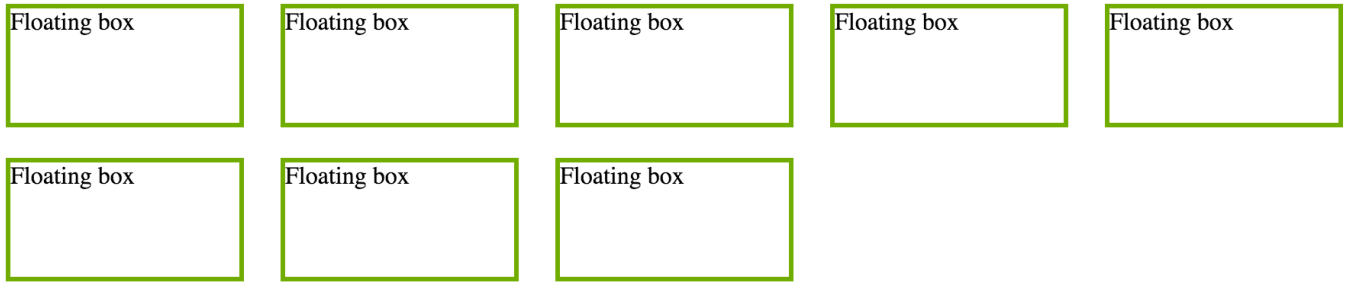
Float Example




```html
<html>
<head>
  <title>My first web page</title>
  <style>
    div {
      border: 3px solid blue;
      width: 1024px;
      height: 128px;
    }
    .float {
      float: right;
    }
    .overflow {
      overflow: scroll;
    }
  </style>
</head>
<body>
  <div>a div block with an image<img class="float" src="bg_image.jpg"></div>
  <p style="clear:right">Style separation</p>
  <div class="overflow">an overflow property with an image
    <img class="float" src="bg_image.jpg"></div>
</body>
</html>
```

Inlineblock

- Inlineblock allows to create a grid of boxes using display property
  - display: inline-block
- Try to resize the page




```html
<!DOCTYPE html>
<html>
<head>
<style>
.floating-box {
    display: inline-block;
    width: 150px;
    height: 75px;
    margin: 10px;
    border: 3px solid #73AD21;
}
</style>
</head>
<body>

<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>
<div class="floating-box">Floating box</div>

</body>
</html>
```

References

- http://www.w3schools.com/css/default.asp - CSS Tutorial